

Skills Summary

Complex Numbers: Complete Solution Sets

Use this reference while completing your worksheet or when studying.

1. Multiplying complex numbers

The one rule: $i^2 = -1$, so $\sqrt{-k} = i\sqrt{k}$ for $k > 0$.

Multiply like binomials, then replace i^2 : $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$.

Powers of i cycle with period 4: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Divide the exponent by 4 and keep the remainder.

Example: Multiply $(3 + 2i)(1 - 4i)$.

Step 1. Two binomials, so distribute all four products — the only special step is turning i^2 back into a real number.

Step 2. $3 \cdot 1 = 3$, $3 \cdot (-4i) = -12i$, $2i \cdot 1 = 2i$, $2i \cdot (-4i) = -8i^2 = +8$. Real: $3 + 8 = 11$; imaginary: $-12i + 2i = -10i$.

$$\Rightarrow (3 + 2i)(1 - 4i) = \mathbf{11 - 10i}$$

Try it: Multiply $(5 + i)(2 - 3i)$.

check: $13 - 11i$

2. Quadratics with complex solutions

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Read the discriminant $\Delta = b^2 - 4ac$ first: $\Delta > 0$ two real solutions; $\Delta = 0$ one repeated real solution; $\Delta < 0$ a *conjugate pair* $p + qi$ and $p - qi$. There is never “no solution” once complex numbers are allowed.

Shortcut: if the variable sits inside a single square, take square roots of both sides instead — and keep both signs.

Example: Solve $x^2 + 2x + 10 = 0$.

Step 1. Nothing multiplies to 10 and adds to 2, so the quadratic formula is the tool; check the discriminant to see what kind of answer to expect.

Step 2. $\Delta = 4 - 40 = -36$, so $\sqrt{\Delta} = 6i$ and $x = \frac{-2 \pm 6i}{2} = -1 \pm 3i$.

$$\Rightarrow \mathbf{x = -1 + 3i \text{ or } x = -1 - 3i}$$

Try it: Solve $x^2 - 6x + 34 = 0$.

check: $5 + 3i$ or $5 - 3i$

3. Testing a proposed solution

Substitute and simplify to one number. A value is a solution exactly when the left side collapses to 0.

Square carefully: $(p + qi)^2 = p^2 + 2pqi + q^2i^2 = (p^2 - q^2) + 2pqi$ — the q^2 term changes sign, and that sign change is the whole check.

Free information: with real coefficients, if $p + qi$ is a solution then $p - qi$ is one too. Never report just one of a pair.

Example: Is $x = 2 - i$ a solution of $x^2 - 4x + 5 = 0$?

Step 1. “Is it a solution?” is settled by substitution, not by re-solving — replace x and simplify.

Step 2. $(2 - i)^2 = 4 - 4i + i^2 = 3 - 4i$; $-4(2 - i) = -8 + 4i$; adding 5 gives $(3 - 8 + 5) + (-4 + 4)i$.
 $\Rightarrow$ the substitution gives **0**, so yes, it is a solution.

Try it: Substitute $x = -1 + 2i$ into $x^2 + 2x + 5$.

check: 0, so it is a solution

4. Reporting the complete solution set

Count first: a degree- n polynomial equation has exactly n solutions, counted with multiplicity. If your list is short, you stopped early.

Route: factor as far as you can over the reals; every leftover $x^2 + k$ contributes the pair $x = \pm i\sqrt{k}$.

- Factor out the common x before dividing by it, or the solution $x = 0$ disappears.
- Only even powers? Factor as a quadratic in x^2 first.

Example: Find every zero of $x^4 - 1$.

Step 1. Degree 4, so four zeros are owed; it is a difference of squares, so factor rather than expand.

Step 2. $x^4 - 1 = (x^2 - 1)(x^2 + 1)$. The first gives $x = 1$ and $x = -1$; the second gives $x^2 = -1$, so $x = \pm i$.

$\Rightarrow$ **$x = 1, -1, i, -i$**

Try it: Find every zero of $x^4 + 5x^2 + 4$.

check: $x = i, -i, 2i, -2i$